

Twentieth International Linguistics Olympiad

Bansko (Bulgaria), 23–29 July 2023

Team Contest Problem

The Murrinh-patha dictionary compiled by Chester S. Street with the help of Gregory Panpawa Mollingin 40 years ago, in 1983, begins as follows:

Murrinh-patha is spoken by approximately 1,100 [Aboriginal people] (as either their first or second language) who live at Port Keats — Wadeye, Northern Territory, 250 kilometres to the south-west of Darwin. A small number of Murrinh-patha speakers also live on nearby cattle stations, and a number live at Kununurra, Western Australia.

At the time of the 2016 census, there were 1,973 native speakers of Murrinh-patha. According to some sources, there are more than 2,500 speakers now. It is one of the few Australian Aboriginal languages whose number of speakers has increased and whose usage has expanded over the past generation.

The dictionary includes English–Murrinh-patha and Murrinh-patha–English sections and is 84 pages long. Sometimes there are pictures in the margins. The entries under the letter M in the second section begin on page 57 and end on page 61. All of these pages are reproduced below, with some minor adjustments and omissions. Further examples from the PhD thesis of Michael James Walsh from 1976 have also been added. English translations of the entries on each page are given after the relevant page, in arbitrary order.

- (a) Match the Murrinh-patha words and phrases with their English equivalents. Each Murrinh-patha dictionary entry has a single English equivalent.
- (b) Restore translation 60-T-54 that was replaced by ***.

(du, m) stands for dual masculine form. (du, f) stands for dual feminine form.

Knowledge of the different species mentioned in the problem is not necessary for solving the problem.

No additional explanation besides the answers is required, nor will be marked.

—*Boris Iomdin, Milena Veneva*

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English text: Boris Iomdin, Milena Veneva.

Good luck!

{ 1	ma		{ 33	MARDABI	bamardabinu
2		ma!	34		
{ 3	MA		35		bangammardabi
4		nanthi perrenemadhaneme pirrine	36		bangamnginthamardabi
{ 5		murrinh wangu nenamanu nukunu-nu	{ 37	MARDARDE	
6		nanthi pi demma dim	38		damnardardenu thurru
{ 7	MABI		39		mukmuk bangammardarde ngem
8		memabinu	{ 40	MARDARDU	
{ 9		memmabi	41		buywardardu nukun
10	marda		42		banmardardu ngem
{ 11	marda		{ 43	MARDALE	
12		marda the mabath	44		damngimardale murrinh thanguu
{ 13	marda	nanthi	{ 45	MARDALILI	
14		marda manganart ngem	46		nanthi thay banmardalili pirrim
{ 15		nanthi	47		bumardalilinu
16	MARDA		{ 48	MARDARL	
17		banhimardanu	49		ngamardarlu
{ 18		bangammarda	50		nanthi tharntu dammardarl wurran
19	MARDA		{ 51	MARDAN	
20		be[nu]mardanu	52		bangimardannu
{ 21		bem[nu]marda	53		bangampunmardan dim
{ 22	MARDA		{ 54	MARDANGKARDU	
23		nganumardanu	55		banhimardangkardunu
{ 24		nganthinnumarda ngarrim	56	MARDANGURRK	
25	MARDA		57		murrinh memmardangurrk dim
{ 26		ngunhimardanu mani?	{ 58	MARDARARTAL	
27		panmarda	59		pangimardarartalnu
{ 28	MARDA		60		damngimardarartal
29		ngunamardanu	{ 61	MARDARR	
{ 30		ngamnarnamarda	62		bemardarnu
31	MARDABAY		63		nanthi bemmardarr nganam
{ 32		panngimardabay dim			

57-T-1	<i>the tree is swaying (in the wind)</i>	57-T-33	<i>I will save</i>
57-T-2	<i>Can I request something from you_{sg}?</i>	57-T-34	<i>I don't know</i>
57-T-3	<i>I desire it [lit. my belly has got it]</i>	57-T-35	<i>it surged</i>
57-T-4	<i>I will hold him to my chest</i>	57-T-36	<i>I gave to him</i>
57-T-5	<i>I will give to you_{sg}</i>	57-T-37	<i>pregnant</i>
57-T-6	verb negator – archaic	57-T-38	<i>I will be satisfied</i>
57-T-7	<i>I met him</i>	57-T-39	<i>she is making a string design</i>
57-T-8	<i>it will surge</i>	57-T-40	<i>Give it to me!</i>
57-T-9	<i>I am out of breath</i>	57-T-41	<i>they are satisfied</i>
57-T-10	<i>I know his thoughts</i>	57-T-42	<i>I will know your_{sg} thoughts [lit. I will see your_{sg} belly]</i>
57-T-11	<i>they (du, f) met him</i>	57-T-43	<i>I rejoiced for him</i>
57-T-12	<i>abdomen (belly), the seat of the emotions</i>	57-T-44	<i>I will have a stomach ache</i>
57-T-13	<i>I am disillusioned</i>		
57-T-14	<i>I rejoiced at the news</i>	57-T-45	<i>to hold/take to one's chest</i>
57-T-15	<i>I requested something from him</i>	57-T-46	<i>to give</i>
57-T-16	<i>I will sway</i>	57-T-47	<i>to be puffed out, to be out of breath</i>
57-T-17	<i>I had a stomach ache</i>	57-T-48	<i>to be disappointed with something, to be disillusioned</i>
57-T-18	<i>I am holding him to my chest (standing)</i>	57-T-49	<i>to request something</i>
57-T-19	<i>the boat is ploughing through (the water)</i>	57-T-50	<i>to know another's thoughts</i>
57-T-20	non-conjugated verb: <i>to give</i>	57-T-51	<i>to mumble</i>
57-T-21	<i>they were passing the thing on/along</i>	57-T-52	<i>to surge (fresh or salt-water)</i>
57-T-22	<i>I will plough the ground</i>	57-T-53	<i>to give to oneself</i>
57-T-23	<i>I will meet him</i>	57-T-54	<i>to rejoice</i>
57-T-24	<i>he is mumbling</i>	57-T-55	<i>to rejoice (at news, etc.)</i>
57-T-25	<i>I will give to myself</i>	57-T-56	<i>to meet</i>
57-T-26	<i>I might get out of breath</i>	57-T-57	<i>to be satisfied</i>
57-T-27	<i>large bamboo (used for spears)</i>	57-T-58	<i>to sway</i>
57-T-28	<i>I gave to myself</i>	57-T-59	<i>to pass on/along; to make string designs with one's hands</i>
57-T-29	<i>you_{sg} will continually know their thoughts</i>	57-T-60	<i>to have a stomach ache</i>
57-T-30	<i>pass the word along to him!</i>	57-T-61	<i>to plough</i>
57-T-31	<i>I will rejoice for him</i>	57-T-62	<i>to know another's thoughts [lit. to see one's belly]</i>
57-T-32	<i>I save things</i>	57-T-63	<i>to save up, to hoard</i>

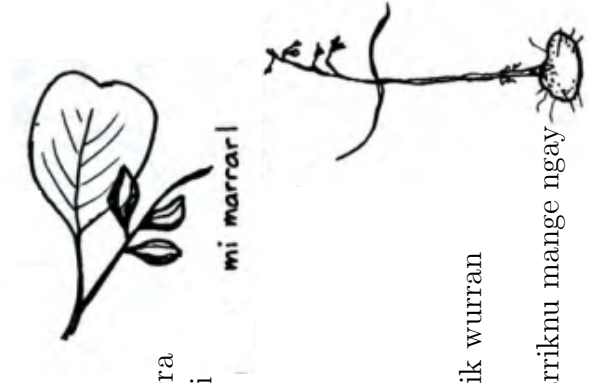
[illegible]

62	mamay	kardu	
{	maminmanga	ku	
{			ma yukuwi ku maminmanga
63			
64			
{	mampurkin	nanthi	
{	mamurt	nanthi	
65		mi	
66			
{	manangka		manangka thipmam
67			
68			
{	manangkaningi	kardu	
69			
70			
{	mani		bangkardu mani
71			
72			
{	manman	ku	
73		nanthi	
74		mi	
75			



58-T-1	<i>message stick</i>	58-T-40	<i>they (du, m) faced each other</i>
58-T-2	<i>thunder and lightning</i>	58-T-41	<i>he was collecting money from them</i>
58-T-3	<i>I will rectify it</i>	58-T-42	<i>unmarried woman</i>
58-T-4	<i>oh, yes, it is a black cormorant</i>	58-T-43	<i>he was washed ashore</i>
58-T-5	<i>didgeridoo</i> (musical instrument)	58-T-44	<i>I continually reprove him</i>
58-T-6	<i>butterfly</i>	58-T-45	<i>I will flash it</i> (operate the projector)
58-T-7	<i>fruit from butterfly tree</i>	58-T-46	<i>left-overs</i>
58-T-8	<i>they (du, f) married</i>	58-T-47	<i>wedge shell</i> (mollusc)
58-T-9	<i>edible fruit from Morinda citrifolia</i>	58-T-48	<i>I collected them from him</i>
58-T-10	<i>he is lamenting for another man</i>	58-T-49	<i>butterfly tree</i>
58-T-11	<i>season — end of the wet season</i>	58-T-50	<i>not</i>
58-T-12	<i>I will move straight through the middle</i>	58-T-51	<i>if it had been me, I would have rectified it</i>
58-T-13	<i>many people had filled up the place instead</i>	58-T-52	<i>praying mantis</i> (insect)
58-T-14	<i>not black</i>	58-T-53	<i>baby girl</i>
58-T-15	<i>the lightning will flash</i>	58-T-54	<i>they (du, f) will marry</i>
58-T-16	<i>water lily</i>	58-T-55	<i>How many praying mantises did you_{sg} see?</i>
58-T-17	<i>young child</i>	58-T-56	<i>heart</i>
58-T-18	<i>and the few men were amazed at all those fish</i>	58-T-57	<i>fruit tree</i>
58-T-19	<i>they (du, m) will face/oppose each other</i>	58-T-58	<i>saver, hoarder</i>
58-T-20	<i>I can have a look</i>	58-T-59	<i>the picture (movie) flashed at us</i> (we saw the picture)
58-T-21	<i>I told him off</i>	58-T-60	<i>he will be washed ashore</i>
58-T-22	<i>women's dance</i>	58-T-61	<i>I flashed it</i> (operated the projector)
58-T-23	<i>season — new leaves appear</i>	58-T-62	<i>I will amaze you_{sg}</i>
58-T-24	<i>Hey! (It is) a wedge shell</i>		
58-T-25	<i>small wallaby</i> (a wallaby is like a kangaroo, but smaller)		
58-T-26	<i>chest</i> (body part)	58-T-63	<i>to be upset, to be troubled</i>
58-T-27	<i>I will be upset</i>	58-T-64	<i>to have hunger pains</i>
58-T-28	<i>I have hunger pains</i>	58-T-65	<i>to be amazed, to marvel</i>
58-T-29	particle that indicates ability to do something	58-T-66	<i>to collect from another</i>
58-T-30	<i>I amazed him</i>	58-T-67	<i>to marry</i> [lit. <i>to put a finger into (a ring)</i>]
58-T-31	<i>black stingray</i> (fish)	58-T-68	<i>to fill up with people (only)</i>
58-T-32	<i>he is upset</i>	58-T-69	<i>to rectify</i>
58-T-33	<i>Morinda citrifolia</i> (tree)	58-T-70	<i>to flash</i> (used of operating a projector)
58-T-34	<i>black cormorant</i> (bird)	58-T-71	<i>to be opposite, to face</i>
58-T-35	<i>crab legs</i>	58-T-72	<i>to wash ashore</i>
58-T-36	<i>the lightning flashed at me</i>	58-T-73	<i>to move straight through the middle; to tell another off, to reprove</i>
58-T-37	<i>edible fruit from nanthi thay malangan</i>	58-T-74	<i>to amaze another</i>
58-T-38	<i>I will have hunger pains</i>	58-T-75	[verb for] <i>lightning flash</i>
58-T-39	<i>unmarried man</i>		

{ 1 2 3	MANPI	me[nu]manpinu ngumem[nu]manpi ngarnam	kardu	{ 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60	MANGKART	memangkartnu memmangkart wurran ku mapu mabibirlnu mere nangimathputh nukun thurru pumengimathpuththa pardi
{ 4 5 6 7 8	manthayway MANTHI	nguymanthinu ngammanthi ku malgaga pammanthawath wurran	nanthi ku	{ 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60	MAPUTH	ku mapu mabibirlnu mere nangimathputh nukun thurru pumengimathpuththa pardi
{ 9 10 11	marnirra marmpi	ngay nganyekum ku marrawarnpam kamayya ku marpi kamayya ku math kamayya	nanthi ku	{ 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60	MARIT	thamul marra marra kanhi
{ 12 13 14 15 16 17 18 19	manhpilyi mange mange mange ... mani	da kura murrinh da nanthi	da mange patha mange kanhi mani	{ 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60	MARRIK	panngimarrik wurran nguy[nu]marriknu mange ngay
{ 20 21 22 23 24	mange numi mange ngala mange ngay mange wurrung purpurk	nanthi nanthi	nanthi nanthi	{ 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60	MART	namart mangaungimart mere menhimartadha manganmarta ngurrammangawurt
{ 25 26 27 28	mangini ... mani mangka mangkamurr	mangini kanhi mani kardu mi	mangini kanhi mani kardu mi	{ 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60	MARTA	mere menhimartadha manganmarta ngurrammangawurt



59-T-1	<i>mother with first-born child</i>	59-T-31	<i>like this</i>
59-T-2	<i>he took it from me</i>	59-T-32	<i>new spear</i>
59-T-3	<i>I didn't understand you_{sg}</i>	59-T-33	<i>tree, used for making wooden hammers and dugout canoes</i>
59-T-4	<i>one who has a speech impediment</i>	59-T-34	<i>fingers</i>
59-T-5	<i>he continually chooses one item</i>	59-T-35	<i>new, now</i>
59-T-6	<i>father's mother, father's mother's sister(s)</i>	59-T-36	<i>an action</i>
59-T-7	<i>I will choose it</i>	59-T-37	<i>I continually understand</i>
59-T-8	<i>thumb, big hand</i>	59-T-38	<i>a good action</i>
59-T-9	<i>here now</i>	59-T-39	<i>cold rain—in dry season</i>
59-T-10	<i>season — showers of rain in the cold season; also eucalyptus flower time</i>	59-T-40	<i>I will help myself</i>
59-T-11	<i>like this</i>	59-T-41	<i>perch (fish)</i>
59-T-12	<i>spider</i>	59-T-42	<i>hand</i>
59-T-13	<i>fruit tree</i>	59-T-43	<i>wild grape</i>
59-T-14	<i>small pied goose (bird)</i>	59-T-44	<i>edible fruit from nanthi thay marrarl</i>
59-T-15	<i>catfish (fish)</i>	59-T-45	<i>five</i>
59-T-16	<i>wooden hammer made from thay maran, used for breaking open mi lala</i>	59-T-46	<i>sign language</i>
59-T-17	<i>I will singe the pied geese</i>	59-T-47	<i>I understood him</i>
59-T-18	<i>I forget: maybe [it was] a pied stilt, maybe a catfish, maybe a perch</i>	59-T-48	<i>I carried it over my shoulder</i>
59-T-19	<i>like, similar</i>	59-T-49	<i>bush potato</i>
59-T-20	<i>pied stilt (bird)</i>	59-T-50	<i>I am going by myself</i>
59-T-21	<i>sky</i>	59-T-51	<i>they were continually disturbing me</i>
59-T-22	<i>by myself [lit. my hand]</i>		
59-T-23	<i>I will trust myself</i>	59-T-52	<i>to learn by observation</i>
59-T-24	<i>don't you_{sg} continually disturb me</i>	59-T-53	<i>to trust</i>
59-T-25	<i>he always carries the wallaby over his shoulder</i>	59-T-54	<i>to help oneself</i>
59-T-26	<i>like, similar</i>	59-T-55	<i>to carry over one's shoulder</i>
59-T-27	<i>we always help each other</i>	59-T-56	<i>to trust oneself</i>
59-T-28	<i>he continually trusts me</i>	59-T-57	<i>to choose, to select (one item)</i>
59-T-29	<i>I will carry it over my shoulder</i>	59-T-58	<i>to understand</i>
59-T-30	<i>you_{sg} took it from him</i>	59-T-59	<i>to take from another</i>
		59-T-60	<i>to disturb from rest/sleep</i>

{ 1 2 3 }	MATHAK		nanthi pungimathaknu ku were pannimathak	{ 32 33 34 }	MENGGAT	panmengkat panhimengkat
4	matharr	nanthi		35	me purrpurrk	nanthi
5	matharr pimut	nanthi		36	mere	
{ 6 7 8 }	MATHBATH		memathbathnu tharra memmathbath ngurran	37	merrk	nanthi
				38		da
				39		ku
9	-mawu			40		nanthi
10			Darwinmawu	41	merrk nguliminki	
11			Kangkarlmawu	{ 42 43 }	METHARDAY	nanthi ngurntin pangankumethardaynintha
12	MAYEPUP		binnamayepup			
13				44	me thingin	nanthi
14	mayi	ku/kardu		45	mi	
15	mayirdarri	ku/kardu		46	mikilik	nanthi
16	mayiyin	ku		47	mikmunurr	kardu
17	me	nanthi		48		
{ 18 19 20 }	MEKAT		medeyi ngem bangamngimekat bengimerrardha tharra	49		kardu palngun mikmunurr kardu nugarn mikmunurr
				50	milinthat	
				51		ku
21				52	milu	ku
22				53	mimpi	nanthi
{ 23 24 25 }	MEKAT		ngumekatnu ku ngurlmiri paumerrarr wurran	54	mirnarthi	kardu
26	merl	ku				
27	MERLMERL		mamngimerlmerl dim			
28						
29	me ngardarda	nanthi				
30			kardu ngardardama			
31	me ngala	nanthi				



60-T-1	<i>I tripped you_{sg}</i>	60-T-29	<i>murderess</i> : actually refers to a woman who by her interaction with several men causes jealousy, and thereby they kill each other off for that woman
60-T-2	verb negator	60-T-30	<i>sickness, illness</i>
60-T-3	<i>I had a dog with me</i>	60-T-31	<i>I am exhausted</i>
60-T-4	<i>the One who abides above</i>	60-T-32	<i>I am carrying it</i>
60-T-5	<i>one month</i>	60-T-33	<i>widow</i>
60-T-6	suffix to abide, to reside	60-T-34	<i>I am hungry</i> [lit. <i>I'm sitting with hunger</i>]
60-T-7	<i>I don't know</i>	60-T-35	<i>person with sore ankle(s)</i>
60-T-8	<i>grass whistling duck</i> (bird)	60-T-36	<i>murderer</i>
60-T-9	<i>toenail</i>	60-T-37	<i>necklace</i> [< Eng. <i>necklace</i>]
60-T-10	<i>big toe, big foot</i>	60-T-38	<i>the wave(s) knocked us off our feet</i>
60-T-11	<i>sandfly</i> (insect)	60-T-39	<i>seaweed</i>
60-T-12	<i>dragon-fly</i> (insect)	60-T-40	<i>he always takes all the fish</i>
60-T-13	non-conjugated verb: <i>to be hungry</i>	60-T-41	<i>foot/feet, track</i>
60-T-14	<i>then I chanced on a sea snake as I went along</i>	60-T-42	<i>I tripped him</i>
60-T-15	<i>widow or widower</i>	60-T-43	<i>widower</i>
60-T-16	<i>full moon</i>	60-T-44	<i>toes</i>
60-T-17	<i>I heard (the story) from him</i>	60-T-45	<i>I tripped/stumbled</i>
60-T-18	<i>I will take everything</i>		
60-T-19	<i>fever</i>		
60-T-20	<i>orphan (or child with just one parent)</i>		
60-T-21	<i>moon</i>		
60-T-22	<i>I will carry it</i>	60-T-46	<i>to take all</i>
60-T-23	<i>one who resides in Darwin</i>	60-T-47	<i>to have with one (things, animals, people)</i>
60-T-24	<i>ankle</i>	60-T-48	<i>to be knocked off one's feet (by waves or strong current)</i>
60-T-25	<i>I will have things with me</i>	60-T-49	<i>to be exhausted</i>
60-T-26	<i>I was continually tripping</i>	60-T-50	<i>to carry a stretcher, etc.</i>
60-T-27	<i>sea snake</i>	60-T-51	<i>to hear (the story) from another</i>
60-T-28	<i>pearly nautilus shell</i> (a nautilus is a type of mollusc)	60-T-52	<i>to trip another</i>
		60-T-53	<i>to trip, to stumble</i>

60-T-54	***
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61-T-1	<i>the thunder is rumbling</i>	61-T-28	<i>speech, language, name(s), places associated with talking or learning, non-Aboriginal songs, stories, legends, news</i>
61-T-2	<i>(place name)</i>	61-T-29	<i>grevillea (plant)</i>
61-T-3	<i>she will dance</i>	61-T-30	<i>old time, long ago, before, etc.</i>
61-T-4	<i>he habitually eats fish</i>	61-T-31	<i>brains</i>
61-T-5	<i>I will speed up the clapsticks</i>	61-T-32	<i>bone</i>
61-T-6	<i>feather(s)</i>	61-T-33	<i>nasty person</i>
61-T-7	<i>old</i>	61-T-34	<i>snake</i>
61-T-8	<i>I will keep the thing for myself</i>	61-T-35	<i>purple shell (mollusc)</i>
61-T-9	<i>thick (blanket, etc.)</i>	61-T-36	<i>a quiet person (i. e. doesn't talk much)</i>
61-T-10	<i>coolibah tree</i>	61-T-37	<i>bony, skinny, lean</i>
61-T-11	<i>the thunder was rumbling</i>	61-T-38	<i>season — dry/cold</i>
61-T-12	<i>he hit the bird</i>	61-T-39	<i>bitter yam</i>
61-T-13	<i>dangerous snake</i>	61-T-40	<i>water lily</i>
61-T-14	<i>I squashed the lice eggs</i>	61-T-41	<i>tail</i>
61-T-15	<i>clapping sticks (used for creating rhythm by clapping the sticks together), firing sticks (used for lighting a fire)</i>	61-T-42	<i>news</i>
61-T-16	<i>correct, pleasant</i>	61-T-43	<i>old thing</i>
61-T-17	<i>I am keeping the thing for myself</i>	61-T-44	<i>lice eggs</i>
61-T-18	<i>brackish water</i>	61-T-45	<i>they were dancing</i>
61-T-19	<i>ornament worn in hair for dancing (made from feathers)</i>	61-T-46	<i>children's talk</i>
61-T-20	<i>good storyteller [lit. beautiful mouth]</i>		
61-T-21	<i>flock pigeon (bird)</i>	61-T-47	<i>to keep something for oneself</i>
61-T-22	<i>baby girl</i>	61-T-48	<i>to dance, women only</i>
61-T-23	<i>I will beautify the top/back of it</i>	61-T-49	<i>to beautify</i>
61-T-24	<i>alcohol</i>	61-T-50	<i>[verb for] thunder rumbling</i>
61-T-25	<i>bird</i>	61-T-51	<i>to eat (more than one thing, or habitually)</i>
61-T-26	<i>water lily</i>	61-T-52	<i>to blink one's eye</i>
61-T-27	<i>dangerous, nasty, cheeky</i>		